

Physics Mock Test

Alternating Current (AC)

Class 12 | UNIT 4 (B)

Marks: 24

Time: 60 Minutes

Syllabus Scope: RMS values, LCR Series Resonance, Phasors, Power in AC Circuits (Wattless Current), and Transformers.

General Instructions: Show all mathematical steps clearly for Section 2. Use $\pi = 3.14$.

Section 1: Multiple Choice Questions (1 Mark Each)

1. Peak and RMS Relation

If the peak value of an alternating voltage is V_0 , then its Root Mean Square (RMS) value is expressed as:

- a) V_0
- b) $V_0 / \sqrt{2}$
- c) $\sqrt{2} \cdot V_0$
- d) $V_0^2 / 2$

2. Purely Resistive Circuit

In a purely resistive AC circuit, the phase difference (ϕ) between the applied voltage and the resulting current is:

- a) 0°
- b) 90°
- c) 180°
- d) 45°

3. Series LCR Impedance

The total opposition (Impedance Z) offered by a series LCR circuit to alternating current is:

- a) $R + X_L + X_C$
- b) $\sqrt{R^2 + (X_L - X_C)^2}$
- c) $X_L - X_C$
- d) $\sqrt{R^2 + X_L^2 + X_C^2}$

4. Condition of Resonance

At the state of electrical resonance in a series LCR circuit, the impedance of the circuit becomes:

- | | |
|------------|-------------------------|
| a) Maximum | b) Minimum (equal to R) |
| c) Zero | d) Infinite |

5. Power Factor Definition

The power factor in an AC circuit is mathematically defined as the ratio of:

- | | |
|---------------------------------|--|
| a) Apparent power to Real power | b) Real (True) power to Apparent power |
| c) Peak current to RMS current | d) Inductive reactance to Capacitive reactance |

6. Ideal Transformers

For an ideal transformer with no energy losses, the electrical power output is always:

- | | |
|-----------------------------|-------------------------------|
| a) Less than power input | b) Equal to power input |
| c) Greater than power input | d) Independent of turns ratio |

Section 2: Numerical & Short-Answer (2-3 Marks Each)

7. Voltage Calculations

An AC voltage source is rated with a peak value of 220 V. Find its Root Mean Square (RMS) value. What is the average value of this voltage over a full cycle?

8. RLC Network Impedance

A series RLC circuit has resistance $R = 10\ \Omega$, Inductance $L = 0.1\ \text{H}$, and Capacitance $C = 100\ \mu\text{F}$. If it is connected to a 50 Hz AC supply, calculate the impedance (Z) of the circuit.

9. Determining Resonance Freq

Calculate the resonant frequency of a series LCR circuit where $L = 0.01\ \text{H}$ and $C = 100\ \mu\text{F}$.

10. Power in AC Network

In an AC circuit, the RMS voltage is 220 V and RMS current is 2 A. If the power factor of the circuit is 0.8, calculate the True Power and the Reactive Power consumed.

11. Transformer Loads

An ideal step-up transformer has 500 turns in the primary and 5000 turns in the secondary coil. If the primary voltage is 220 V, calculate the secondary voltage. If the load on secondary draws 2 A, what is the primary current?

12. Full Circuit Analysis

In a series LCR circuit, $V_{\text{rms}} = 200 \text{ V}$, $R = 20 \Omega$, $X_L = 30 \Omega$, and $X_C = 10 \Omega$. Find:

- (i) The RMS current (I)
- (ii) The power factor ($\cos \phi$)
- (iii) Total power consumed.

Answer Key & Solutions

1. b) $V_0 / \sqrt{2}$

Effective or RMS voltage is $0.707 \times$ Peak voltage.

2. a) 0°

In a resistor, voltage and current are in same phase.

3. b) $\sqrt{[R^2 + (X_L - X_C)^2]}$

Derived from the phasor triangle.

4. b) Minimum (equal to R)

As $X_L = X_C$, the reactive parts cancel each other.

5. b) Real (True) power to Apparent power

Power Factor = $\cos \phi = P_{\text{true}} / P_{\text{apparent}}$

6. b) Equal to power input

Ideal efficiency is 100% (Energy is conserved).

7. 155.56 V | Zero

$V_{\text{rms}} = 220 / 1.414 \approx 155.56$ V. Average of a full sine wave is always 0.

8. $\approx 10.01 \Omega$

$X_L = 2\pi fL = 31.4\Omega$. $X_C = 1/(2\pi fC) = 31.83\Omega$. $Z = \sqrt{[10^2 + (31.4 - 31.83)^2]} \approx 10.01 \Omega$.

9. ≈ 159.24 Hz

$f_r = 1 / (2\pi\sqrt{LC}) = 1 / (2 \times 3.14 \times \sqrt{10^{-6}}) = 1 / (6.28 \times 10^{-3})$.

10. 352 W | 264 VAR

True P = $VI \cos \phi = 220 \times 2 \times 0.8 = 352$ W. Reactive P = $VI \sin \phi = 220 \times 2 \times 0.6 = 264$ VAR.

11. $V_s = 2200$ V | $I_p = 20$ A

$V_s = 220 \times (5000/500) = 2200$ V. Since power is conserved ($V_p I_p = V_s I_s$), $I_p = (2200 \times 2) / 220 = 20$ A.

12. (i) 7.07 A | (ii) 0.707 | (iii) 1000 W

$Z = \sqrt{[20^2 + 20^2]} = 20\sqrt{2} \approx 28.28 \Omega$.

(i) $I = 200/28.28 = 7.07 \text{ A}$. (ii) $\cos \phi = R/Z = 20/28.28 = 0.707$. (iii) $P = 200 \times 7.07 \times 0.707 = 1000 \text{ W}$.

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